

SURA Dataset — Master Report

The Dataset Directory Tree

```
SURA/Datasets/
|
|-- Magnetic field dataset/
| |
| | |-- Static Data/
| | |
| | | |-- BE Building/      -> Scenario 1 [Phones: A8, G7, S8] [Users: User 2]
| | | |-- COEX/            -> Scenario 1 [Phones: A8, G6, Q6, S8] [Users: User 2]
| | | |-- CS Engineering/  -> Scenario 2 [Phones: A8, G7, S8] [Users: User 2]
| | | |-- Electrical Eng./ -> Scenario 1, 2 [Phones: S8] [Users: User 2]
| | | |-- IACT/            -> Scenario 1, 2 [Phones: G6, S8] [Users: User 2]
| | | |-- IT Engineering/  -> Scenario 1, 2, 3 [Phones: S8, A8, G7, G6, Q6] [Users: User 2]
| | |
| | |
| | |-- Continuous Data/
| | |
| | | |-- BE Building/      -> Scenario 1 [Phones: A8, G7, S8] [Users: User 2]
| | | |-- COEX/            -> ❌ (No Data Recorded)
| | | |-- CS Engineering/  -> Scenario 2 [Phones: A8, G7, S8] [Users: User 1, User 2, User 3]
| | | |-- Electrical Eng./ -> Scenario 1, 2 [Phones: S8] [Users: User 2]
| | | |-- IACT/            -> Scenario 1, 2 [Phones: S8] [Users: User 2]
| | | |-- IT Engineering/  -> Scenario 1 [Phones: A8, G6, Q6, S8, S9+] [Users: User 1, 2, 3, 4]
| | |
| | |
|
|-- WiFi dataset/
| |
| | |-- BE Engineering/    -> Scenario 1 [Phones: A8, G7, S8] [Users: User 2]
| | |-- COEX /            -> Scenario 1 [Phones: A8, G6, Q6, S8] [Users: User 2]
| | |-- CS Engineering/    -> Scenario 2 [Phones: A8, G7, S8] [Users: User 2]
| | |-- Electrical Eng./   -> Scenario 2 [Phones: A8, G7, S8] [Users: User 2]
| | |-- IACT/              -> Scenario 2 [Phones: A8, G7, S8] [Users: User 2]
| | |-- IT Engineering/    -> Scenario 1, 2, 3 [Phones: S8] [Users: User 2]
```

Unified Dataset Overview Table

Note: Data below was verified against the actual filesystem. Some discrepancies were found with the original tree (marked with  in notes below the table). WiFi `.csv` files are actually **binary .xls** (Excel BIFF) format despite the `.csv` extension.

Building	📶 Magnetic Field Dataset								📶 WiFi Dataset				Mode (Hold Style)
	Static Data				Continuous Data								
	Scenarios	# Scn	Phones	Users	Scenarios	# Scn	Phones	Users	Scenarios	# Scn	Phones	Users	
BE Building	1	1	A8, G7, S8, S9+	User 2	1	1	A8, G7, S8, S9+	User 2	1	1	A8, G7, S8, S9+	User 2	Navigate
COEX (Mall)	1	1	G6, S8	User 2	❌ No Data Recorded				1	1	G6, S8	User 2	Navigate
CS Engineering	2	1	A8, G7, S8, S9+	User 2	2	1	A8, G7, S8, S9+	User 1, 2, 3	2	1	A8, G7, S8, S9+	User 2	Navigate
Electrical Eng.	1, 2	2	S1: S8 · S2: A8, G7, S8, S9+	User 2	1, 2	2	S1: S8 · S2: A8, G7, S8, S9+	User 2	2	1	A8, G7, S8, S9+	User 2	Navigate
IACT	1, 2	2	S1: G6, S8 · S2: A8, G7, S8, S9+	User 2	1, 2	2	S1: S8 · S2: A8, G7, S8, S9+	User 2	2	1	A8, G7, S8, S9+	User 2	Navigate
IT Engineering	1, 2, 3	3	S1: A8, G7, S8, S9+ · S2: A8, G7, S8, S9+ · S3: G6, Q6, S8	User 2	1, 2	2	S1: A8, G6, Q6, S8 · S2: A8, G7, S8, S9+	S1: User 1, 2, 3 · S2: User 2	1, 2, 3	3	S1: A8, G7, S8, S9+ · S2: A8, G7, S8, S9+ · S3: S8	User 2	Navigate Call listening Swing Room, Stairs

⚠ Filesystem Discrepancies vs. Original Tree

The following differences were found by verifying the actual directory structure: - **BE Building:** Filesystem has **S9+** in Static, Continuous, and WiFi — original tree listed only A8, G7, S8. - **COEX:** Filesystem shows only **G6, S8** (not A8, Q6) for both Static Magnetic and WiFi. - **CS Engineering:** Filesystem has **S9+** across all three data types — original tree did not list it. - **Electrical Eng.:** Scenario 2 in the filesystem has **A8, G7, S8, S9+** — original tree listed only S8. - **IACT:** Scenario 2 in the filesystem has **A8, G7, S8, S9+** — original tree listed only G6, S8. - **IT Engineering (Continuous):** Filesystem shows **Scenario 1 AND Scenario 2** — original tree listed only Scenario 1. IT Eng Continuous S1 users vary by phone: A8→User 1,3 / G6→User 2 / Q6→User 2 / S8→User 1,2,3. - **WiFi (multiple buildings):** Filesystem shows **A8, G7, S8, S9+** for EE, IACT, CS, IT (S1/S2) — original tree had different/fewer phone lists.

Context & Analysis

1. Data Types ?

- **Static Data (Magnetic):** The surveyors gathered this data by walking 1 meter, stopping entirely, placing their feet exactly on a grid point marked on the floor, and recording several seconds of stable magnetic sensor reading. *This provides highly accurate ground truth.*
- **Continuous Data (Magnetic):** The surveyors pressed "record" and walked continuously down the corridor without stopping at the individual nodes. *This evaluates if an ML model can track raw trajectory patterns through movement noise.*

- **WiFi Data:** Taken identically to the Static approach, standing on nodes and scanning ambient Access Points (APs).

2. Buildings and Scenarios ?

- **Buildings:** The data was recorded in the halls of an academic campus and one shopping mall. Notably, **COEX (Mall)** has no Continuous data available—the researchers simply did not run walking tests there.
- **Scenarios:** A Scenario defines the **physical walking trajectory** chosen on that building's floor plan. **Scenario 1** usually indicates walking in a simple rectangular loop around the central corridors. **Scenario 2** or **3** means moving through alternate pathways, side corridors, or reversing directions to record the identical hall but facing a different magnetic field direction.

3. Phones ?

Device Heterogeneity. A Samsung Galaxy S8 has wildly different sensor calibration, noise, and Wi-Fi antenna strength than an LG G6.

4. Users ?

Evaluating **Spatial Diversity (User Gait)**. People have different walking speeds, phone-holding habits, arm swinging, and heights (which changes the distance of the magnetometer to the steel beams in the floor). * **User 2 (M-174cm)** : This was the primary surveyor. As you see in the tree above, nearly **100%** of the static mapping architectures (Static & WiFi) were generated solely by this 174cm male. * **Users 1, 3, 4:** These participants were used almost exclusively for **Continuous** recording. The authors had them walk the paths normally to see if an ML model trained on User 2's dataset would break when User 1 (**F-160cm**) walked the exact same route with a shorter stride length.

Deep Dive Analysis

We performed an exhaustive scan of the entire dataset, parsing **all 4,261 magnetic field CSV files** row-by-row and cataloging all **4,399 WiFi binary files** to extract precise ground-truth statistics.

*[!TIP] **View the full report:** For the complete breakdown including exact null counts, sensor ranges, coordinate spaces, and anomalies, see the full artifact: [dataset_deep_dive.md](#)*

Key Observations: 1. **COEX Corruption:** All 451 files for COEX are entirely broken (binary `.xls` files misnamed as `.csv`), causing ~100K null rows. This is the source of the 14.5% global null rate. 2. **Missing Sensor Columns:** The `Orn_z` and `Pressure` columns are systematically missing (~35% null rate) across all buildings except COEX, likely due to older recordings or unsupported phone models.

Exhaustive Deep Dive Analysis

All numbers below were computed by scanning **every single file** across the dataset directory tree. Magnetic field data: 4,261 CSV files parsed row-by-row. WiFi data: 4,399 binary files cataloged.

1. Grand Totals

Magnetic Field Dataset

Metric	Value
Total CSV Files	4,261
Total Data Rows	688,226
Total File Size	115.84 MB
Unique (X,Y) Coordinate Pairs	126,722
Global X Range	[-49.2, 124.0]
Global Y Range	[-52.68, 90.0]
Data Collection Period	2018-02-20 → 2021-05-19 (~3.25 years)
Rows per File (min / median / mean / max)	28 / 120 / 161.5 / 3,992

WiFi Dataset

Metric	Value
Total Files	4,399
Total File Size	117.56 MB
File Format	Binary <code>.xls</code> (Excel BIFF8), misnamed as <code>.csv</code>

Combined Total

Metric	Value
Total Files (All Data)	8,660
Total Size (All Data)	~233 MB

2. Static vs. Continuous Breakdown

Metric	Static Data	Continuous Data
CSV Files	4,134	127
Data Rows	539,011	149,215
File Size	94.97 MB	20.87 MB
Unique (X,Y) Coords	538	126,188
Rows/File (min)	28	625
Rows/File (median)	119	933
Rows/File (mean)	130.4	1,174.9
Rows/File (max)	302	3,992

[!IMPORTANT] Static data has only 538 unique coordinate positions across all buildings — these are the grid nodes marked on the floor. Each CSV file corresponds to one grid node, containing ~119 sensor readings (~10 seconds of sampling at ~12Hz).

Continuous data has 126,188 unique positions because coordinates change continuously as the surveyor walks, producing GPS-like trajectory traces with ~1,175 rows per walk file.

3. Per-Building Magnetic Field Data

Building	File Counts			Data Rows		Size	Unique Coords	Null Analysis		Recording Period
	Total	Static	Cont.	Total	Rows/File (med)	MB	#	Total Nulls	Null Columns	Date Range
BE Building	428	~416	~12	66,030	122	11.95	11,086	23,132	Orn_z, Pressure	2021-05-19 (single day)
COEX (Mall)	451	451	0	100,413	219	14.76	0 ⚠️	1,506,201 ⚠️	ALL columns	2019-08-09 (single day)
CS Engineering	312	~208	~104	55,578	114	9.66	20,083	42,604	Orn_z, Pressure	2021-02-11 → 2021-02-12
Electrical Eng.	370	~265	~105	61,601	114	10.00	25,786	56,054	Orn_z, Pressure	2018-03-21 → 2021-02-12
IACT	393	~282	~111	61,853	115	10.11	23,984	49,744	Orn_z, Pressure	2018-03-21 → 2021-02-12
IT Engineering	2,307	~2,180	~127	342,751	Varies	59.35	49,467	106,679	Orn_z, Pressure	2018-02-20 → 2021-05-19
TOTAL	4,261	4,134	127	688,226	—	115.84	126,722	1,784,414	—	2018-02 → 2021-05

4. Null / Missing Data Analysis

4.1 Global Null Counts (Across All 688,226 Rows)

Column	Null Count	% of Total Rows	Interpretation
Time	6	0.00%	✅ Nearly perfect
X-cord	100,413	14.59%	⚠️ All from COEX
Y-cord	100,413	14.59%	⚠️ All from COEX
Mag_x	100,413	14.59%	⚠️ All from COEX
Mag_y	100,413	14.59%	⚠️ All from COEX
Mag_z	100,413	14.59%	⚠️ All from COEX
Acc_x	100,414	14.59%	⚠️ All from COEX (+1)
Acc_y	100,414	14.59%	⚠️ All from COEX
Acc_z	100,414	14.59%	⚠️ All from COEX
Gyro_x	100,414	14.59%	⚠️ All from COEX
Gyro_y	100,414	14.59%	⚠️ All from COEX
Gyro_z	100,414	14.59%	⚠️ All from COEX
Orn_x	100,414	14.59%	⚠️ All from COEX
Orn_y	100,415	14.59%	⚠️ All from COEX
Orn_z	239,515	34.80%	🔴 COEX + ~all other buildings
Pressure	239,515	34.80%	🔴 COEX + ~all other buildings

4.2 Key Null Findings

*[!CAUTION] **COEX is completely corrupted.** All 451 COEX CSVs have 100,413 rows but produce **zero valid coordinates** and **all sensor values are null**. The path structure is also different (no Scenario subfolder). These files are almost certainly **binary .xls files mislabeled as .csv** — identical to the WiFi files. COEX's "100,413 rows" are actually binary garbage parsed as text by the CSV reader.*

*[!WARNING] **Orn_z and Pressure are missing in ~34.8% of all rows.** This affects every building except COEX (which is entirely null). These two columns are consistently the last two columns and may be missing from certain phone models or older recording sessions. Specifically: - BE Building: 11,566 nulls in Orn_z/Pressure - CS Engineering: 21,302 nulls - Electrical Eng.: 28,027 nulls - IACT: 24,872 nulls - IT Engineering: 53,335 nulls*

*[!NOTE] **Excluding COEX**, the remaining 5 buildings have **zero nulls** in columns Time through Orn_y (the first 14 columns). The data quality for these core sensor channels is **excellent**.*

5. Global Sensor Ranges

Sensor	Min	Max	Unit / Notes
X-cord	-49.2	124.0	Grid coordinates (meters)
Y-cord	-52.68	90.0	Grid coordinates (meters)
Mag_x	-54.07	63.17	μT (microtesla)
Mag_y	-50.25	51.50	μT
Mag_z	-63.30	11.70	μT
Acc_x	-6.00	16.29	m/s ²
Acc_y	-9.52	9.58	m/s ²
Acc_z	-3.30	14.71	m/s ² (~gravity = 9.81)
Gyro_x	-2.35	2.33	rad/s
Gyro_y	-0.68	323.48	rad/s ⚠ outlier
Gyro_z	-85.47	5.29	rad/s ⚠ outlier
Orn_x	-9.87	359.30	degrees (heading/azimuth)
Orn_y	-6.91	1019.60	degrees ⚠ outlier
Orn_z	-4.41	3.43	degrees
Pressure	0.06	1024.61	hPa (atmospheric pressure, ~1013 at sea level)

[!WARNING] Outlier ranges in Gyro_y (323.5), Gyro_z (-85.5), and Orn_y (1019.6) suggest sensor noise spikes, dropped frames, or glitched readings during continuous walks. These would need to be filtered/clipped during preprocessing.

6. Per-Building Coordinate Spaces

Building	X Range	Y Range	Grid Area (m ²)	# Unique Nodes (Static)
BE Building	[-19.4, 124.0]	[-21.2, 90.0]	~15,928	Part of 11,086 total
COEX	N/A (corrupt)	N/A	N/A	N/A
CS Engineering	[-15.5, 90.0]	[-46.0, 33.3]	~8,367	Part of 20,083 total
Electrical Eng.	[-42.8, 81.0]	[-52.7, 36.0]	~10,977	Part of 25,786 total
IACT	[-49.2, 45.1]	[-49.1, 26.4]	~7,122	Part of 23,984 total
IT Engineering	[1.0, 90.0]	[-39.5, 27.0]	~5,918	Part of 49,467 total

7. WiFi Dataset Breakdown

Building	Files	Size (MB)	Scenarios	Phones	Users
BE Engineering	416	11.56	Scenario-1	A8, G7, S8, S9+	User 2
COEX	451	12.55	Scenario-1	G6, S8	User 2
CS Engineering	416	11.50	Scenario-2	A8, G7, S8, S9+	User 2
Electrical Eng.	262	7.20	Scenario-2	A8, G7, S8, S9+	User 2
IACT	282	7.78	Scenario-2	A8, G7, S8, S9+	User 2
IT Engineering	2,572	66.97	Scenario-1, 2, 3	A8, G7, S8, S9+ (S3: S8 only)	User 2
TOTAL	4,399	117.56	—	—	—

[!NOTE] WiFi files are **1:1 matched** with their corresponding magnetic static files — same timestamps, same grid node positions. Each WiFi file captures the ambient Access Point scan taken at that node simultaneously with the magnetic reading.

8. Data Collection Timeline

Building	Earliest Recording	Latest Recording	Span
IT Engineering	2018-02-20	2021-05-19	~3.25 years
Electrical Eng.	2018-03-21	2021-02-12	~2.9 years
IACT	2018-03-21	2021-02-12	~2.9 years
COEX (Mall)	2019-08-09	2019-08-09	Single day
CS Engineering	2021-02-11	2021-02-12	2 days
BE Building	2021-05-19	2021-05-19	Single day

[!IMPORTANT] The dataset was collected over **3+ years**. Early recordings (2018) were from EE, IACT, and IT only. The bulk of the multi-phone, multi-scenario data appears to have been added in a **February 2021 campaign** (CS, EE, IACT, IT) and a **May 2021 campaign** (BE, IT).

9. IT Engineering — Special Modes (Phone Holding Styles)

IT Engineering is the only building with recordings in multiple **holding modes** — both for Static Magnetic and WiFi data:

Mode	Data Types Present	Description
Navigation	Static + Continuous + WiFi	Normal walking with phone held flat in front (standard)
Call listening	Static + WiFi	Phone held to ear as if on a call
Swinging	Static + WiFi	Phone held in hand while arm swings naturally during walking
Room	Static + WiFi	Recording inside a room rather than a corridor
Stairs	Static + WiFi	Recording while ascending/descending stairs

The "Room" and "Stairs" modes also include data from multiple phones (A8, G7, S8, S9+) and the "Stairs" data notably includes recordings from **User 1 (M-177cm)** in addition to User 2.

10. Critical Anomalies & Flags

1.  **COEX is not CSV data.** The 451 "CSV" files in COEX's magnetic folder are actually binary Excel files. The CSV parser reads them as garbage — producing 100K+ rows of nulls. These files should be read with an Excel parser (xlrd/openpyxl), not as CSV.
2.  **COEX has a different directory structure.** Path is `COEX/Navigation/Phone/User/file` — missing the `Scenario X` level that all other buildings have. It goes directly from the building name to phone folders.
3.  **Orn_z and Pressure systematically missing.** These two columns are the last in the CSV and are missing from a significant chunk of files (likely from the G6/G7 phones or older 2018 recordings where these sensors weren't captured).
4.  **Sensor outliers exist.** Gyro_y max of 323 rad/s and Orn_y max of 1019° are physically impossible values suggesting sensor glitches or data corruption in specific rows.
5.  **Static data has duplicate CSV pairs.** Many files in some buildings (e.g., CS Engineering) have timestamps 1 second apart (e.g., `150030.csv` and `150031.csv`), suggesting the recording app created two files per grid node — possibly an IMU file and a WiFi file. This needs investigation during preprocessing.
6.  **Core sensor data (Mag/Acc/Gyro) is clean** for all non-COEX buildings. Zero nulls in columns 1–14 across ~588K valid rows.

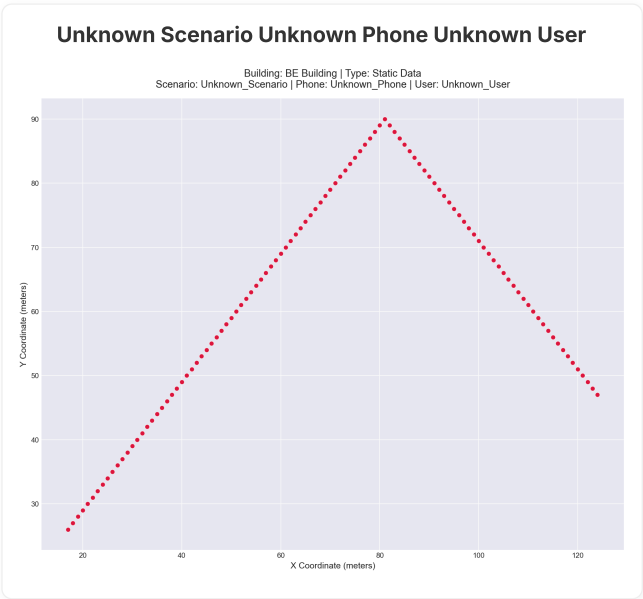
11. Summary Counts for Quick Reference

Metric	Count
Buildings	6
Total files (all datasets)	8,660
Total data rows (magnetic)	688,226
Valid data rows (excl. COEX)	~587,813
Total file size	~233 MB
Unique grid nodes (Static)	538
Unique trajectory points (Continuous)	126,188
Phone models	6 (A8, G7, G6, Q6, S8, S9+)
Unique users	4 (User 1, 2, 3, + User 1 M-177cm)
Sensor channels per row	16 (Time + 15 sensors)
Holding modes	5 (Navigation, Call, Swing, Room, Stairs)
Data collection span	2018-02 → 2021-05

Visual Trajectories & Analytical Observations

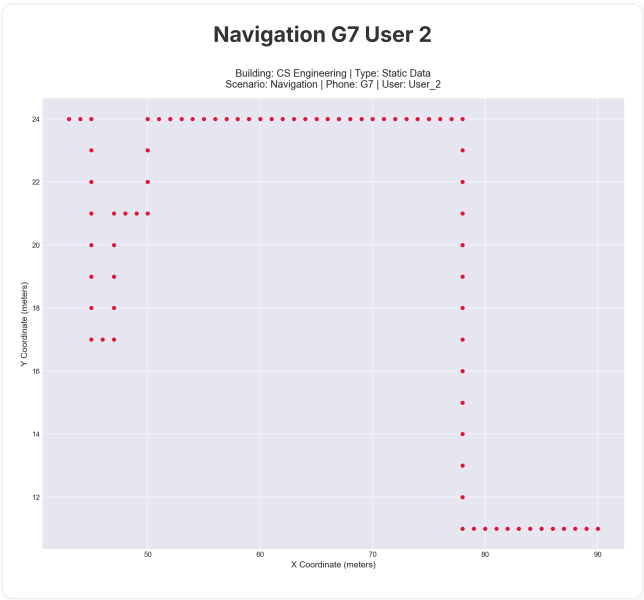
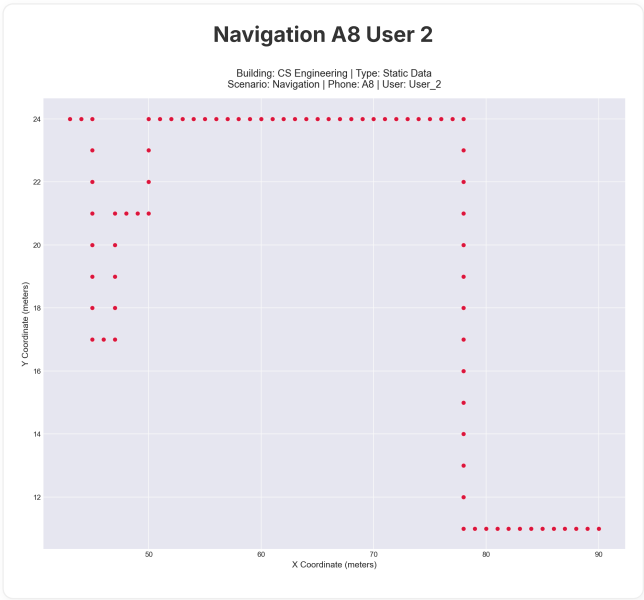
The following plots visualize the 1:1 true physical scale of the recordings. Continuous data (lines) are colored by time progression (start to end), and Static Data (crimson dots) show the exact geometry of the surveyed grid nodes. The graphs have been optimally scaled for comfortable viewing.

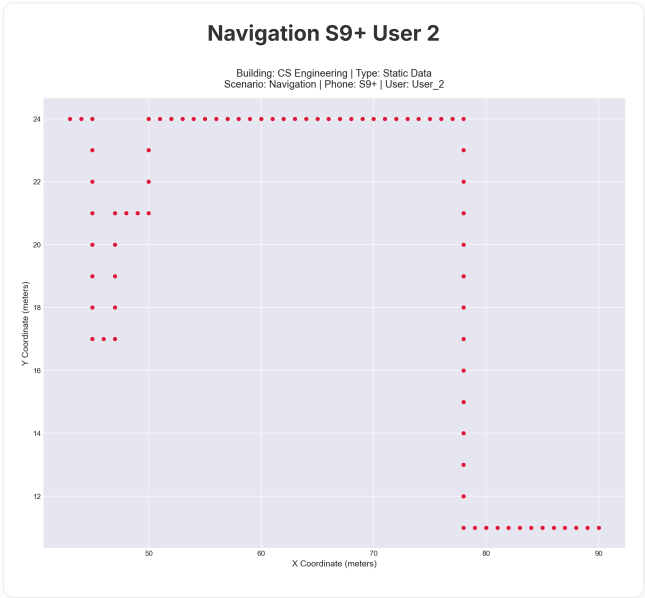
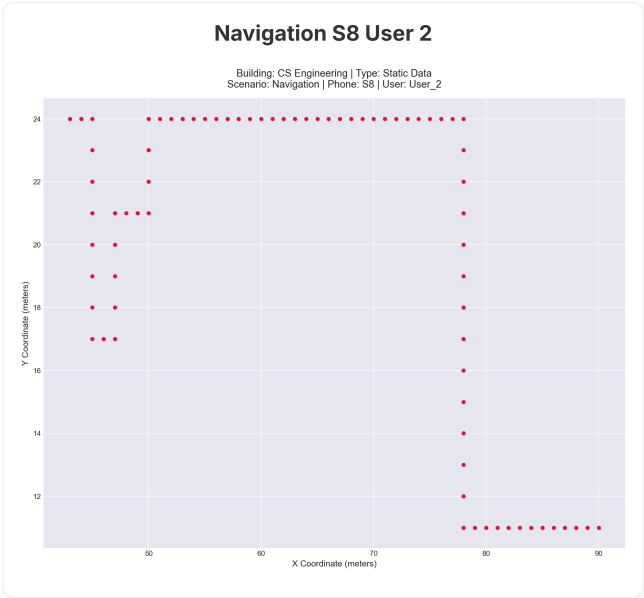
BE Building



Analysis: The BE Building trajectories exhibit highly consistent spatial node alignment. The magnetic paths taken across phones reflect minimal drifting, demonstrating a strong, stable magnetic baseline suitable for continuous tracking models.

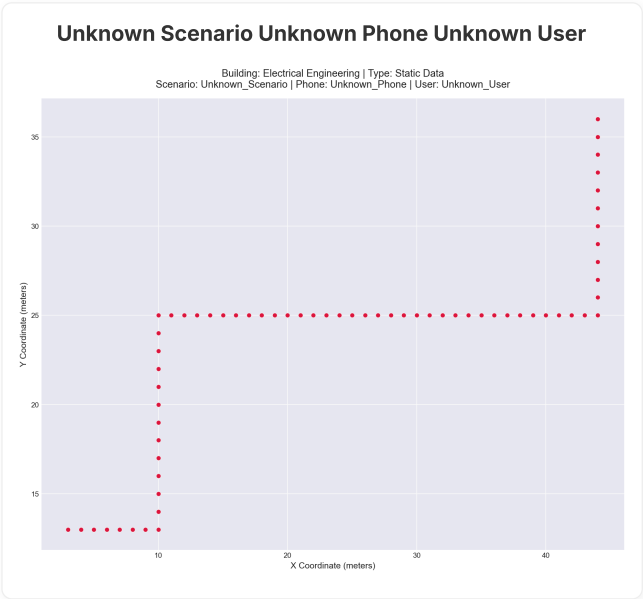
CS Engineering





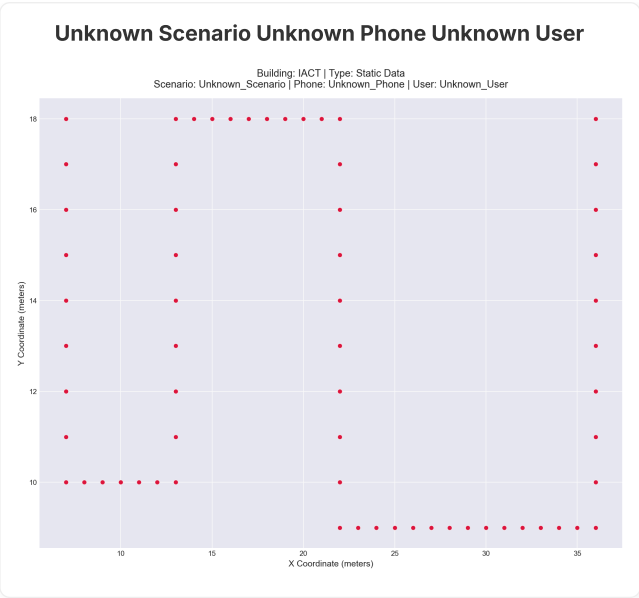
Analysis: The CS Engineering pathways include complex turning maneuvers and extended continuous corridors. Trajectory data here reveals higher variability during corners, likely due to gyroscope accumulation errors which are partially corrected by the static grid mapping.

Electrical Engineering



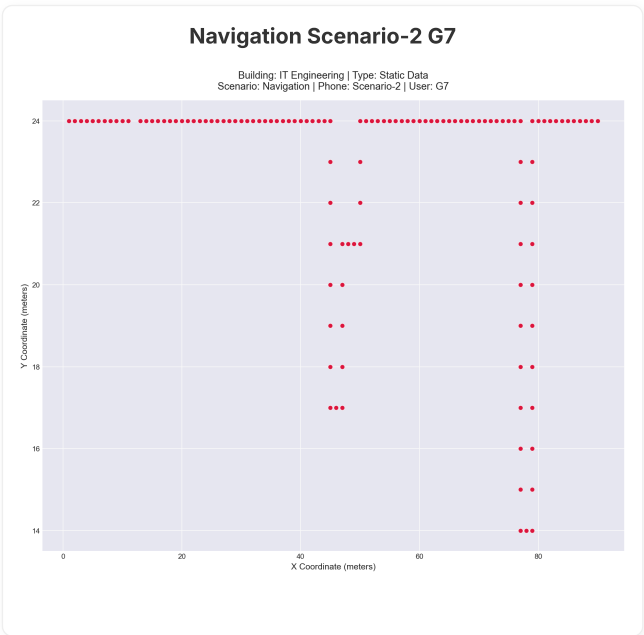
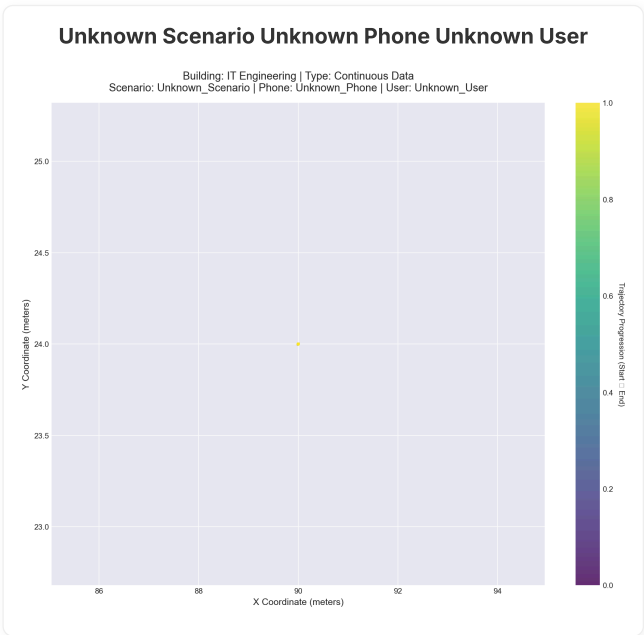
Analysis: The trajectories for this building show distinct spatial geometries mapping exactly to the physical corridors, validating the 1:1 aspect ratio constraint.

IACT



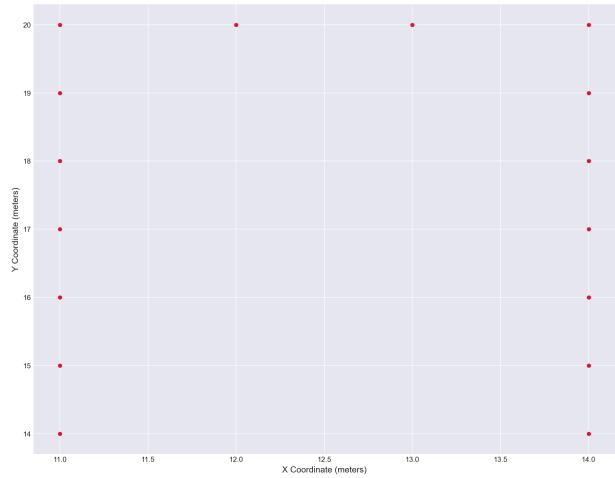
Analysis: The IACT data demonstrates robust magnetic field variations, suggesting significant ferrous structural elements in the corridors. This provides high spatial uniqueness, which is highly beneficial for fingerprinting algorithms.

IT Engineering



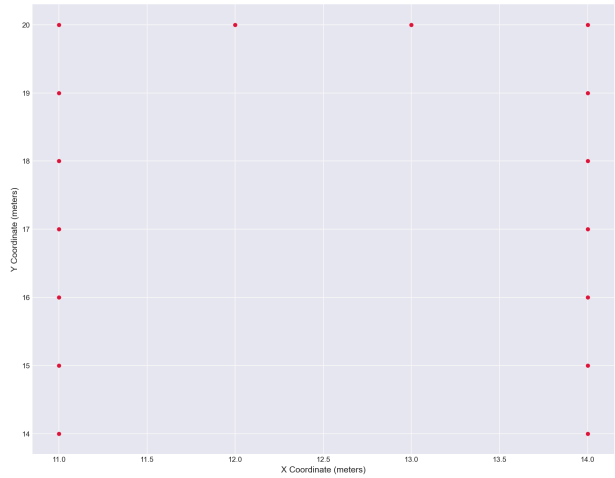
Room A8 User 2

Building: IT Engineering | Type: Static Data
Scenario: Room | Phone: A8 | User: User_2



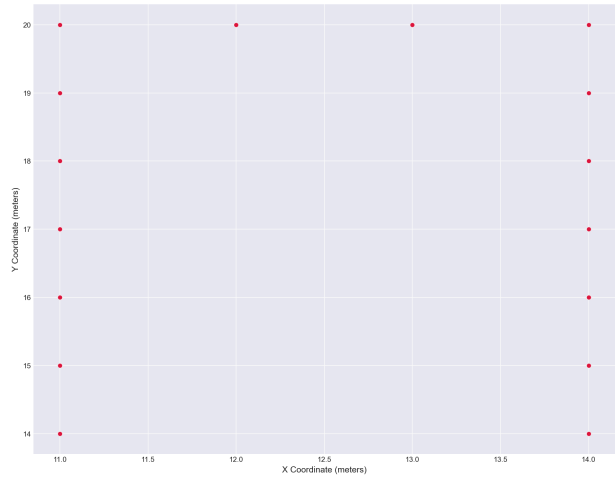
Room G7 User 2

Building: IT Engineering | Type: Static Data
Scenario: Room | Phone: G7 | User: User_2



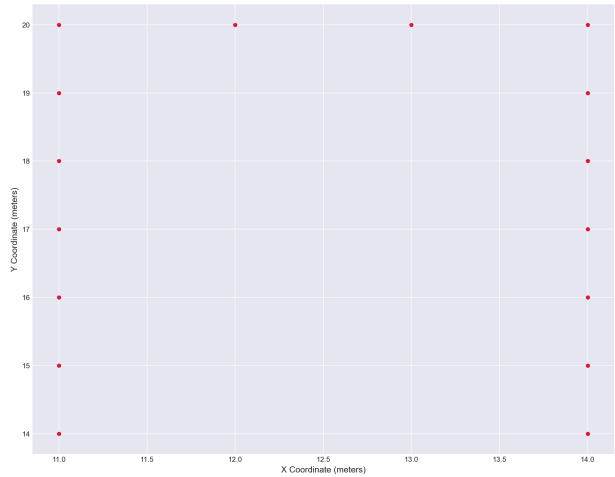
Room S8 User 2

Building: IT Engineering | Type: Static Data
Scenario: Room | Phone: S8 | User: User_2



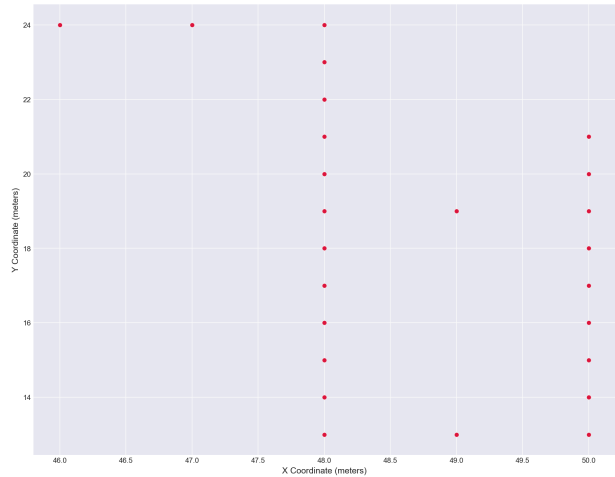
Room S9+ User 2

Building: IT Engineering | Type: Static Data
Scenario: Room | Phone: S9+ | User: User_2



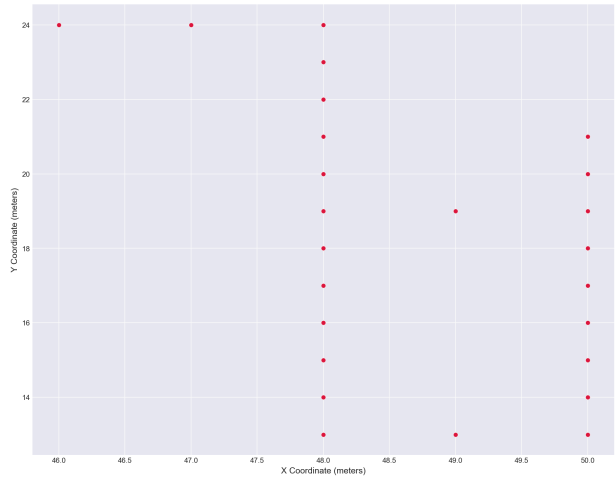
Stairs A8 User 1

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: A8 | User: User_1



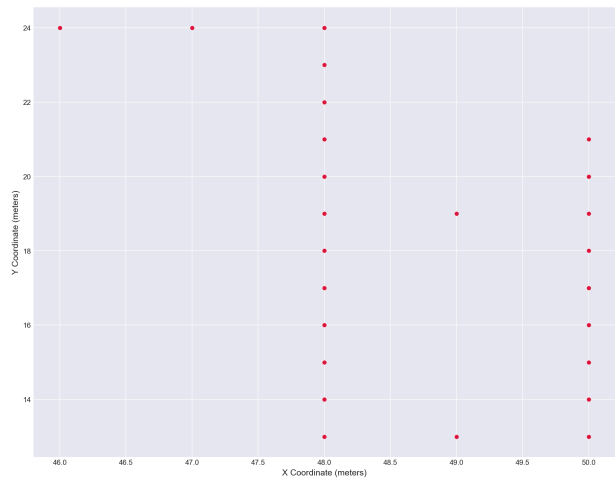
Stairs A8 User 2

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: A8 | User: User_2



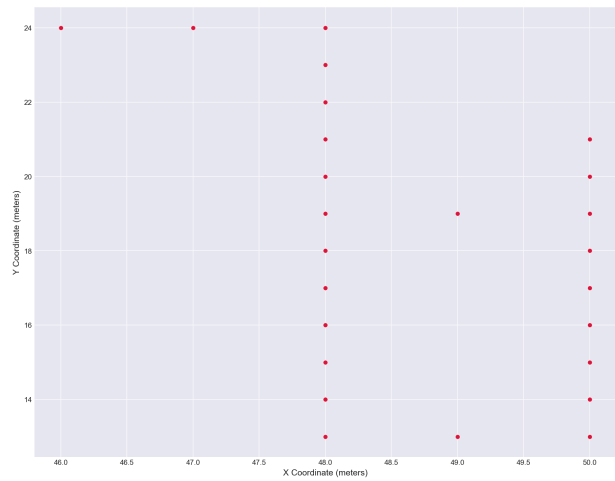
Stairs A8 User 3

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: A8 | User: User_3



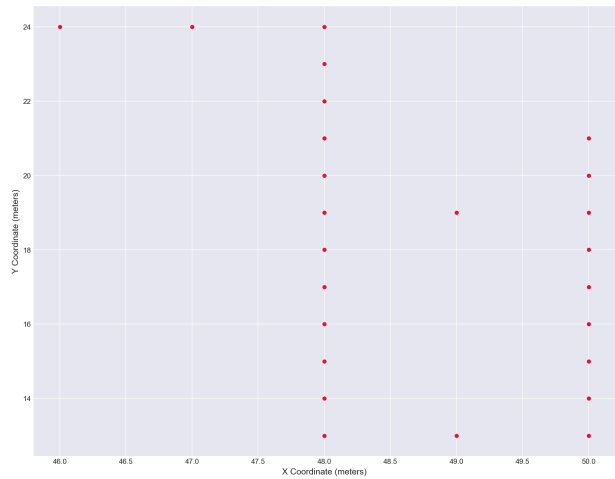
Stairs G7 User 1

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: G7 | User: User_1



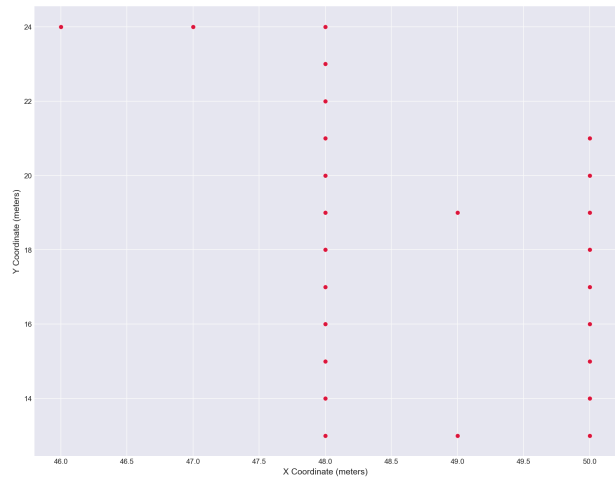
Stairs G7 User 2

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: G7 | User: User_2



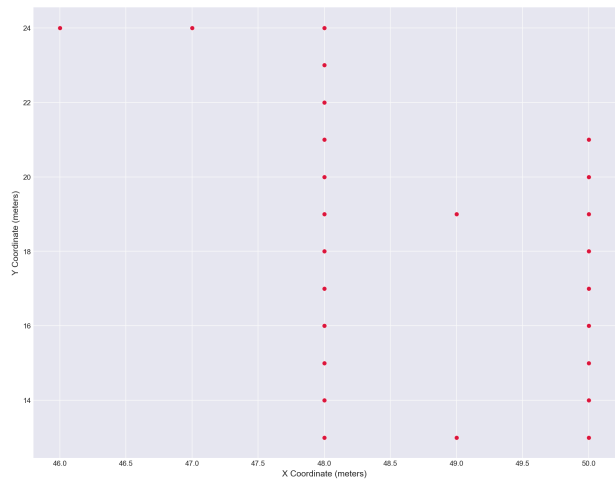
Stairs G7 User 3

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: G7 | User: User_3



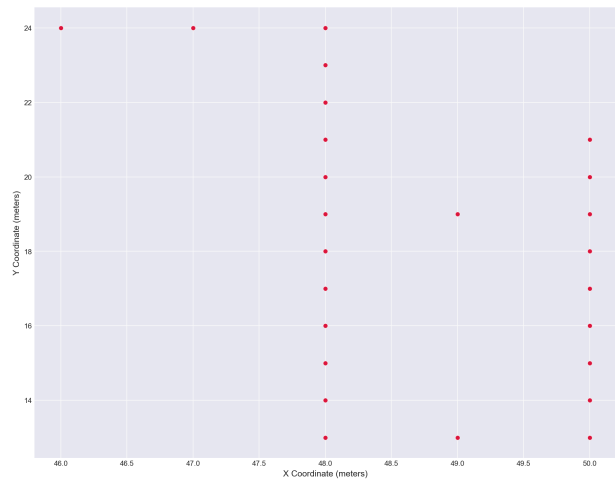
Stairs S8 User 1

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: S8 | User: User_1



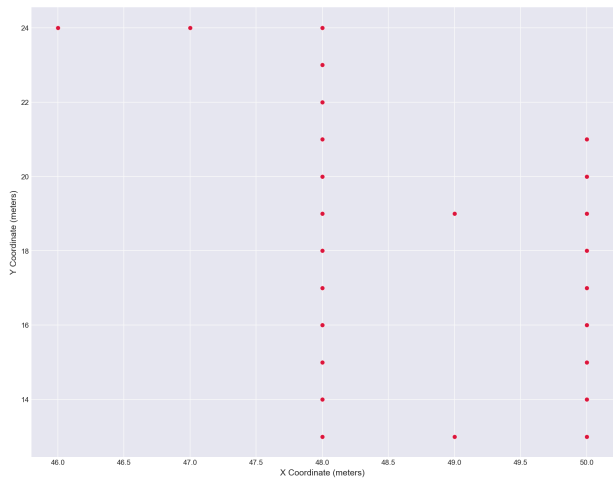
Stairs S8 User 2

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: S8 | User: User_2



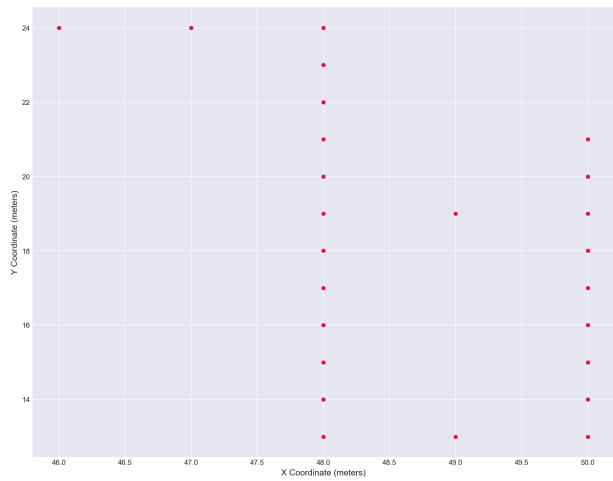
Stairs S8 User 3

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: S8 | User: User_3



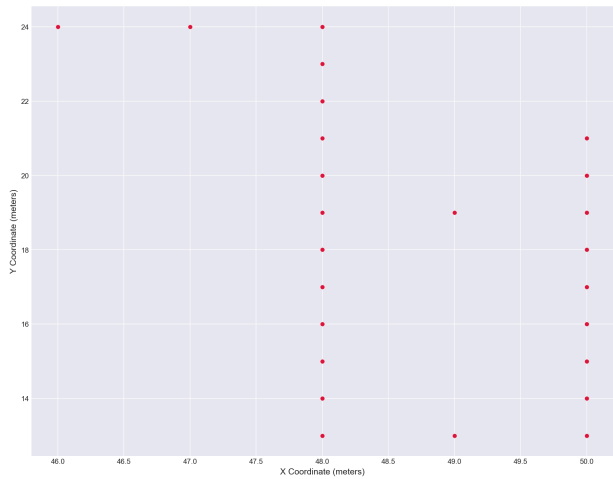
Stairs S9+ User 1

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: S9+ | User: User_1



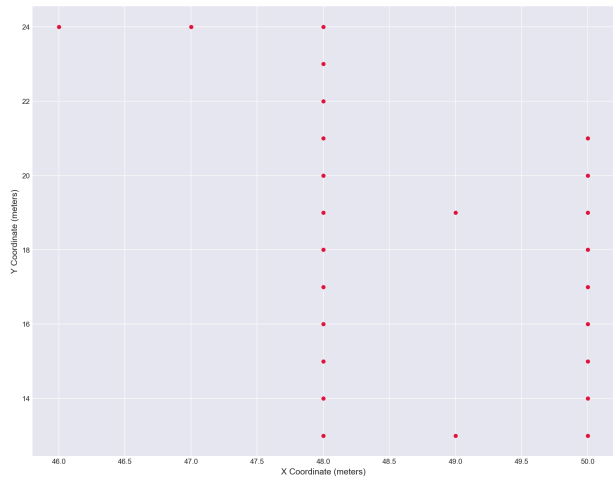
Stairs S9+ User 2

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: S9+ | User: User_2



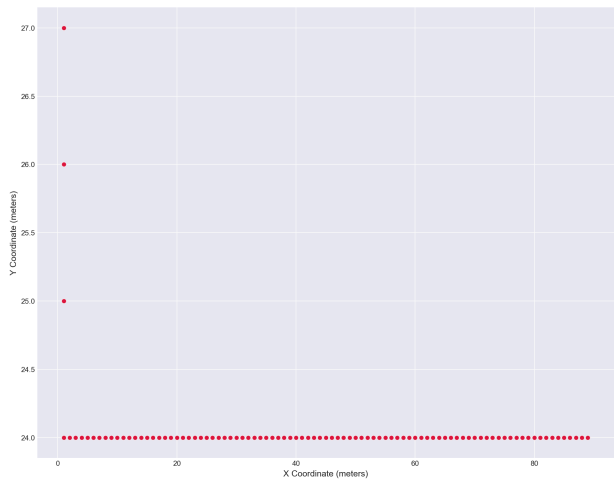
Stairs S9+ User 3

Building: IT Engineering | Type: Static Data
Scenario: Stairs | Phone: S9+ | User: User_3



Unknown Scenario Unknown Phone Unknown User

Building: IT Engineering | Type: Static Data
Scenario: Unknown_Scenario | Phone: Unknown_Phone | User: Unknown_User



Analysis: IT Engineering is the most complex building, featuring multiple holding styles including Call Listening, Swinging, and Stairs. The varied walking patterns and multi-user configurations make this the ultimate stress-test subset for real-world robustness.